# Investigating the Performance of Protocols Within Smart Oceans and their Created Mobile Networks

COMP4032 Advanced Computer Networks Project Report

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# Contents

# Chapter 1

# Introduction

This experiment compares the performance of single-copy routing protocols against a flooding based approach, and evaluates their adaption to a scaled opportunistic network. This project focuses on a simulated smart ocean created from marine wildlife tracking.

Within these networks, tagged wildlife act as sensor nodes that exist in a self-organising topology. The collected data is propagated towards the surface sink nodes via a multi-hop route [1]. This can be analysed as an intermittently connected mobile network which requires delay tolerant routing to achieve successful message delivery. There must be consideration on the energy consumption of the nodes due to the difficulties of recharging or replacing their batteries and the limited capacity available [1]. This highlights the importance of choosing an efficient delivery method, that is able to adapt with network scalability, to minimise transmission of redundant data and overcome the high cost of underwater communications.

The 'Spray and Wait' protocol has been used to explore efficiency of the propagation of multiple copies of data packets, while placing limits on packet replication to reduce congestion [2]. The success of this will be compared against the use of the 'SimBet' protocol, which instead directs a single copy of the packet towards a node that is believed to be more likely to bring a successful delivery [3].

## 1.1 Mobile Ad hoc DTN Networks

Delay Tolerant Networks have been built to address networking scenarios that work among environments with high disruptions and long disconnection periods. This may be caused by nodes with mobility capabilities or data transfer across extreme distances, such as interplanetary communications or deep ocean networks, which can cause inevitable periods of long latency delay [4].

Traditional TCP/IP internet protocols would fail in these cases, as these schemes rely on the assumption that an end-to-end path exists between the sender and receiver at the point of message dissemination [2]. This is not always true within real-world scenarios - even in the case of a static network, intermittent contacts may occur due to a decrease in quality of the communication channel [4]. When available, the mobility of the nodes must be exploited through a "store and carry" process: the node stores the received bundle within its available buffer until an appropriate link to the next hop becomes available, chosen considering characteristics such as work load and buffer capacity [2].

## 1.2 Chosen Simulator

The ONE Simulator v1.6.0 (Opportunistic Network Environment simulator) [5] is used to implement this scenario, illustrate the message passing between nodes, and evaluate the performance of two routing protocols: Spray and Wait, and SimBet. This software has been chosen because of its ability to generate mobility traces among custom nodes and model external traces upon collected real-life data. The variety of statistical-reports are used to evaluate the success of message passing and allow for a multi-criteria comparison of the two routing methods chosen, this is discussed further in Chapter 4.

The Spray and Wait implementation has been provided within the original ONE Simulator source code [6], while the SimBet algorithm has been taken from an open source extension developed by the community [7].

# Chapter 2

# Chosen Protocols

Basic flooding protocols, such as Epidemic Rooting, are likely to bring successful delivery in DTNs but come with a high overhead as nodes spread their knowledge throughout the network. By doing this, there exists many redundant copies of the message, creating competition for the available buffer space of a node. This overloading of the nodes result in a high number of dropped packets, and thus necessary retransmissions, which introduces further network traffic. In order to avoid this, more restrictive message transmission must be implemented [2].

## 2.1 Spray and Wait Approach

Within the paper "Spray and Wait: An Efficient Routing Scheme for Intermittently Connected Mobile Networks" [2], the authors propose an improved routing schema, Spray and Wait, for communication within these intermittent networks. This is introduced as being able to outperform the basic flooding alternatives when evaluated in terms of average latency and the length of transmission path taken.

Unlike the basic replication-based algorithms, Spray and Wait is able to place limits on the number of message copies that are transmitted through the network, by setting the bounding variable L. This allows the key issues of flooding-based protocols to be avoided. This protocol begins with a 'spray phase' where the node forwards a message copy to L distinct nodes, these may be the first nodes encountered or may be selected based on heuristics. After each transmission, the attribute L is decremented until there is only a single copy to send. At this point, it enters its 'wait phase' and acts as a direct transmission towards the destination [2]. The authors [2] discuss the research conducted on the proposed protocol, supporting their hypothesis that Spray and Wait performs well within a scalable network. They have found that this achieves delivery in fewer hops and shorter delays, compared to other flooding-based algorithms, for both low and high load scenarios.

Our simulation explores this scalability of Spray and Wait, and compares performance to an alternative protocol SimBet. At the time that "Spray and Wait: An Efficient Routing Scheme for Intermittently Connected Mobile Networks" was presented, the SimBet routing algorithm had not yet been proposed. Throughout this discussion, these protocols will be analysed to compare how they adapt within changing scenarios to keep a high ratio of success and efficiency.

## 2.2 SimBet Routing

In order to increase the success of message dissemination, deciding the next hop in a forwarding algorithm should preference nodes that have frequent communication with the destination address. This is based on the assumption that nodes within the same group communicate with each other and that nodes that have interacted before will do so again [3].

The SimBet algorithm proposed by Daly and Haahr [8] is a novel approach that can determine the most appropriate forwarding path while considering social characteristics of the neighbouring nodes and its similarity to the destination node. From the research conducted by Hsu and Helmy [9], these encounters can be used to construct a 'connected relationship graph' that allows for a social analysis of Freeman's graph-theory measures, such as the strength of nodes' degree and betweenness centrality [10], and the nodes high in these measures should be preferred. This allows social importance to be determined based on direct and indirect encounters. Using this, the data bundle can be routed towards a 'more central' node that will increase the number of 'useful connections' among the other nodes in the network - ones that will increase the probability of finding a suitable relay node that can reach the destination.

The use of social metrics allows for the avoidance of redundant information exchange: messages will not be forwarded towards nodes with low similarity or low importance, as these are unlikely to have interest in the message and will not propagate closer towards its destination. As discussed by Daly and Haar [8], this approach increases in energy efficiency compared to flood-based methods, since it maintains only a single copy of each message. This brings a further reduction in resource overheads and can be compared to the extreme version of the Spray and Wait approach with L=1, which acts as an inefficient direct transmission. This is particularly important within the analysed scenario as the communication links established between sub-marine nodes can be very resource-intensive and may exist for short periods of time.

# Chapter 3

# Experiment Design

This project considers an alternative approach to the retrieval of data collected by wearable sensors. This creates a network between tagged organisms to transport data from inaccessible environments and take advantage of the wider ecology of the ocean.

## 3.1 Scenario Motivation

The tracking of animals in the ocean poses different challenges to those on land, as they are often outside the reach of satellites. Wearable sensors, such as GPS tags, are commonly used to capture the location of the animal once it breaches the surface of the ocean. Additional sensors are often used to measure trends within the environment, such as temperature, pressure, oxygen content, and bacterial activity. This method of data collection is dependent on the frequency of visits to the water's surface for the required transmission time, making for inefficient monitoring of animals that stay in deeper waters [11].

For these creatures, alternative methods have been explored. For example, the use of purpose-built infrastructure allows location tracking to take place entirely beneath the water's surface. Acoustic telemetry can be used to record the presence of an animal as it passes through known positions. This is highly dependent on the presence of these acoustic receivers; without them there are unavoidable gaps in the recorded path [11]. Some wildlife organisations, such as MarAllliance use wearable 'pop-up' tags store collected sensor data on the animal until a decided time frame has passed, at which point the tag releases and floats to the surface for collection. A manned-visit is needed to retrieve the tag and the data it carries. In order to collect future information about the animal, re-attachment of the tag is necessary [12]. Alternatively, a 'Smart Ocean' allows the sharing of data between organisms and can propagate the information within satellite range.

## 3.2 Scenario Design

This experiment simulates data transmission within a smart ocean network, emulating the movement of marine animals at different ocean depths. This assumes that animals have been equipped with sensors to collect data about themselves and their surroundings and must be routed towards the surface. Due to the high-mobility of the animals, there will be no guaranteed end-to-end route at the time of message dissemination, and contacts within the created network are unpredictable [2].

Success requires a delay tolerant routing approach, a 'direct transmission' algorithm is not applicable in this scenario, since the node carrying the data may never travel sufficiently close to the surface to successfully deliver the packets. Instead, a multi-hop process must be implemented. The collected data can be forwarded along existing links that may occur between other passing organisms and stored until the next appropriate connection is made [2]. By allowing the creation of these links between organisms, we encourage the sharing of data between species with differing movement patterns. Therefore, there is a high probability that the data will be brought within the communication range of the satellites. This will significantly increase the knowledge collected from the ocean and provides a path towards organisms that would otherwise be unreachable.

This scenario could include additional node groups outside of marine wildlife. Data may be shared with static nodes, such as purpose built infrastructure similar to acoustic telemetry tracking [12]. There may also be intentional nodes with predictable or known movement paths, such as boats, divers, and other man-made controllable instances.

When designing the simulation, its important to take into consideration the wider context of the scenario. Within the discussion "Underwater Internet of Things in Smart Ocean: System Architecture and Open Issues" [1], Qiu et al explore ongoing research into underwater node communication and how this differs from the standard Internet of Things - radio channels are unlikely to be available and the use of acoustic telemetry has much lower efficiency and reliability than typical surface communications. A combination of wireless communication technologies can be used to allow for efficient underwater transmission of data. Within this paper, we will abstract away from this problem and assume basic communication is possible between the nodes in order to focus analysis on the routing of the packets.

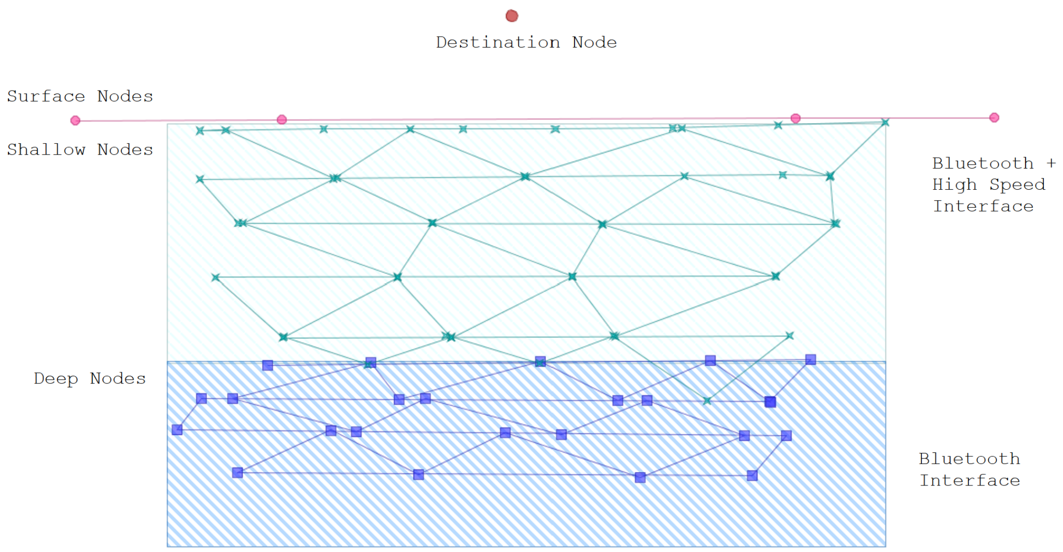



Figure 3.1: Map used for the simulation, with two regions for the shallow and deep nodes.

## 3.3 Simulation Set Up

The movement patterns of the nodes have been estimated to emulate different marine animals recorded in OCEARCH tracking data. A 2D simulation has been used to create a space abstraction that places focus on the network created between animals at varying depths within the smart ocean, rather than on their larger behaviour patterns. The axes have been placed on the depth and longitude positions of the organisms, and the nodes have been separated into four groups, as shown within Figure 3.1.

The first group of nodes are confined to the deepest region of the scenario, to represent marine animals that stay deep beneath the water's surface - such as the Dogfish or Greenland sharks. These deep nodes generate the network traffic and forwards the packets towards the shallower nodes. The second node group models the behaviour of organisms that spend time in shallow waters and will often dive deeper to search for food. This means they will likely make contact with a deeper node to receive the data packet and can act as relay nodes.

Additional nodes on the surface of the water may act as a bridge to the final destination, these are intentional nodes which may represent fishing boats or other infrastructure. The final node group models the satellites as the destination for all network traffic. For the purposes of our simulation, these last two node groups will be implemented as static nodes.

Future exploration may use an external data trace to map real data that has been gathered from tracking ocean wildlife. The nodes within the simulation can be modeled based on real movement patterns, for example using tracking data from an organisation such as OCEARCH shown in Figure 3.2. Within the time limits of the project, a suitable trace could not be accessed that provided the depth of the marine animals along with their tracked locations.

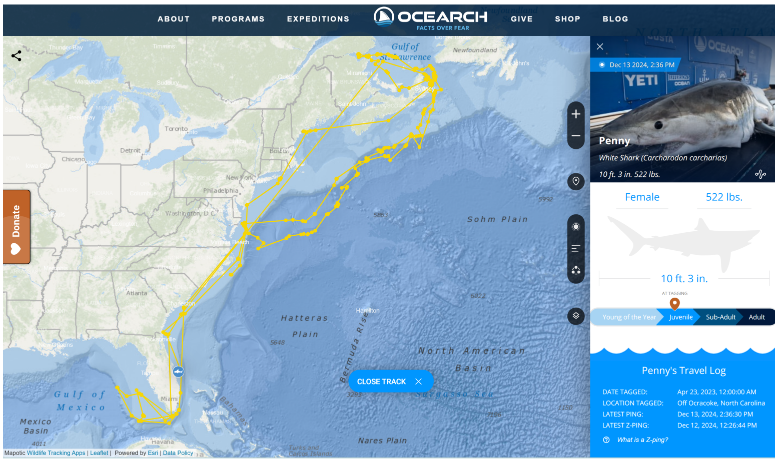



Figure 3.2: Example real-life tracking data of a shark from OCEARCH's database.

## 3.4 Experiment Criteria

Similarly to the research conducted within the Spray and Wait proposal [2], this experiment will explore the performance of the discussed protocols through a comparison of the average delay, and the number of transmissions needed, to successfully deliver data packets within the proposed scenario.

As a secondary investigation, the scalability of each protocol will be analysed through adjustments to the number of nodes within the network. Figure 3.3 shows example network layouts in the ONE simulator. This will explore the link between the size of the network and the number of hops necessary for successful delivery. Since each node may act as a sink, this creates a high rise in network traffic and contention for nodes limited buffer spaces. This may cause both edge and node level congestion. However, having more nodes increases the likelihood that a link will be created with a node that will visit the destination frequently.

(a) A dense 'shallow region' and a sparse 'deep region'.

(b) A sparse 'shallow region' and a dense 'deep region'.

Figure 3.3: An example timestep within two of the experiments, showing the simulation set up for different node populations.

# Chapter 4

# Performance Evaluation

Within the ONE simulator, the classes 'MessageStatsReport', 'MessageDelayReport', 'CreatedMessages', and 'DeliveredMessages' are used to collect information from each experiment and evaluate performance of the delivery protocols. These provide details such as the routes taken by the successful delivery of a packet, the average overhead, latency, and hop count.

## 4.1 Probability of Delivery

The performance of the protocols are evaluated across different simulation configurations to explore how they respond to scaling of the network. The populations of the node groups are modified independently to allow separate investigations into the density of the deep source nodes and the shallow nodes used to relay data. Figures 3.3a and 3.3b show the scenario with examples of dense and sparse regions.

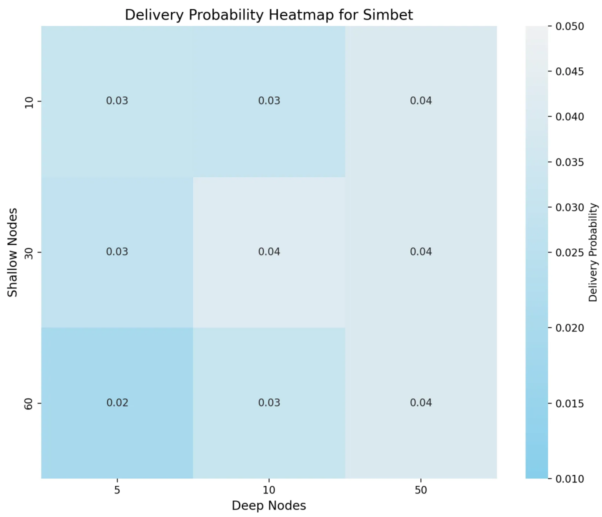



Figure 4.1: A heatmap visualising how SimBet adapts when altering the density of nodes within each region of the map.

The use of the SimBet protocol provided consistently lower probability of deliveries, these are shown in Figure 4.1. The probability of successful message delivery was found to decrease as each region in the network became more sparse. The darker areas in the heatmap show that a low population of deep nodes creates a low delivery probability, which is likely due to the fact that a lower number of nodes occupying the same space reduces the number of possible connections. This reduces the chance that the SimBet router is able to find a connection that will progress the packet closer to its destination. Similarly, a low population of shallow nodes has caused a low delivery rate, as it is less likely that the node will come into contact with a surface node.

A very dense number of deep nodes has given a large delivery probability, with respect to the other trails. Within this simulated scenario, a constant number of message events are generated, therefore more nodes are available and there are a higher number of successful connections to forward the message towards. Within a real life scenario this may have a lesser impact on probability success as each node may be transferring its own data, rather than acting solely as a relay node. In this case, having a greater number of nodes that generate the data traffic may cause congestion throughout the network, which would lead to packet loss and a reduction in successful deliveries.

In contrast, Spray and Wait has been shown to be much more effective under this metric, the results of this are given within the heatmap in Figure 4.2. From our experiment, the scenario with the greatest probability of delivery is the case where the shallow region of nodes is the most dense. This is as the nodes within this region act as data mules between the deeper source nodes and the destination, the flooding aspect of the protocol can take advantage of this by spreading copies of the message and increasing the chance that it is carried to the top of the node's region to be transferred to a surface node.

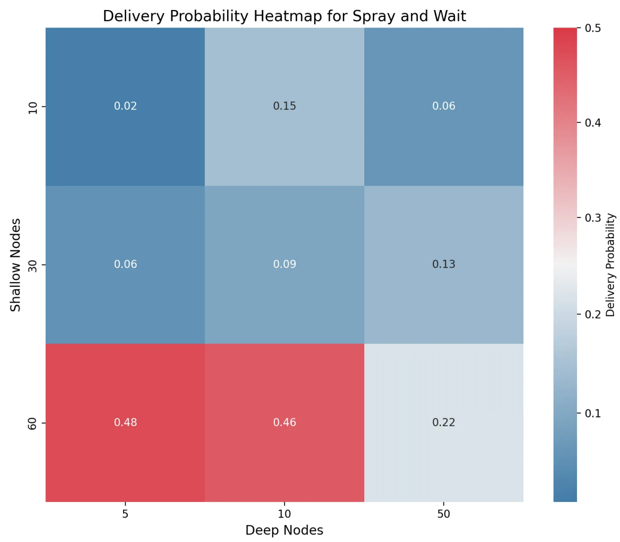



Figure 4.2: A heatmap visualising how Spray and Wait adapts when altering the density of nodes within each region of the map.

## 4.2 Number of Hops

The Spray and Wait protocol has been shown to be the most effective in terms of the number of transmission hops required to reach the destination. This metric remains consistent when increasing the total population of nodes in the scenario. However, SimBet reacts poorly to a highly populated network, requiring twice the number of data transfers to route a message compared to a comparably sparser population. This can be seen in Figure 4.3.

When more nodes are available, the flooding aspect of Spray and Wait can exploit the available connections and propagate multiple message copies; increasing the probability of a low-hop delivery [2]. Comparatively, SimBet is more selective and sends only a single copy to its chosen contact [8] - if these selection conditions are not properly aligned with the scenario this may cause the message to be routed with negative progress.

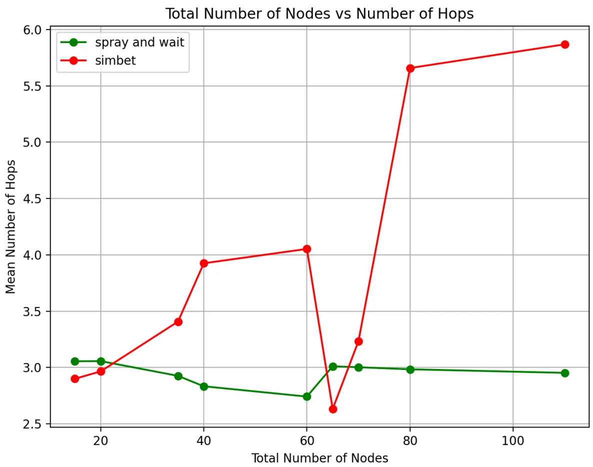



Figure 4.3: A plot visualising the average number of hops for each network scale.

When choosing a protocol within a real world application of this scenario, priority can be placed on either performance metric: minimising the number of hops on a successful route versus restricting the number of redundant message copies throughout the network.

Although Spray and Wait results in a shorter delivery path, when measured by hops, it also largely increases network traffic. If a suitable cap is not placed on packet replication, end-to-end congestion may occur both within the network and node's buffer space - leading to kiss if data. Within an underwater environment there is additional motivation to minismise traffic, and propagation paths, due to the cost if of data transfer [1].

## 4.3 In-Network and Node-Level Latency

In support of the observed trends, the Spray and Wait protocol has been found to be most effective with a scaled network population, as seen in Figure 4.4. When propagating with a replication-based protocol, a smaller, more-sparse network has produced the highest latency. This is since fewer transmission paths are created due to the limited number of node connections.

Larger distances between the nodes may also result in a higher network latency, this is due to the natural delay between connection events and the restrictive buffers that cause queuing delays. This has a less severe impact within a larger network as there is greater redundancy and better reachability among nodes; this increases the potential routes the packet can take [4]. This allows for routing with a more efficient latency, as the message is likely to be successful along the shortest path first.

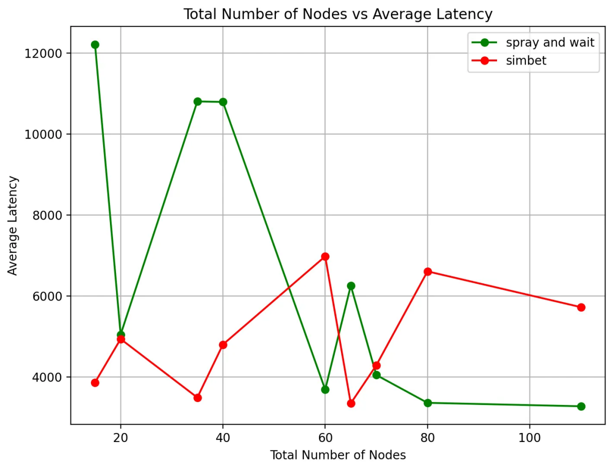
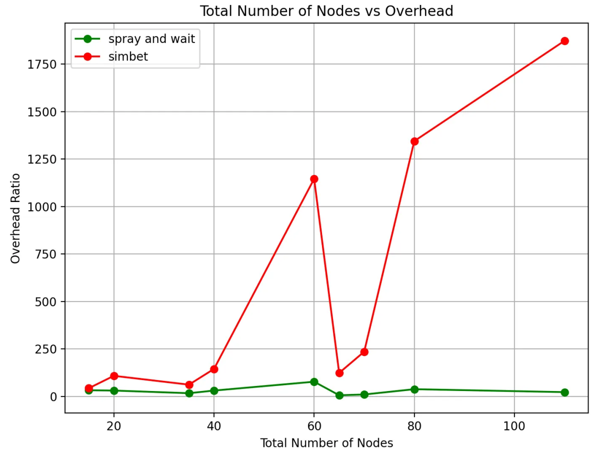


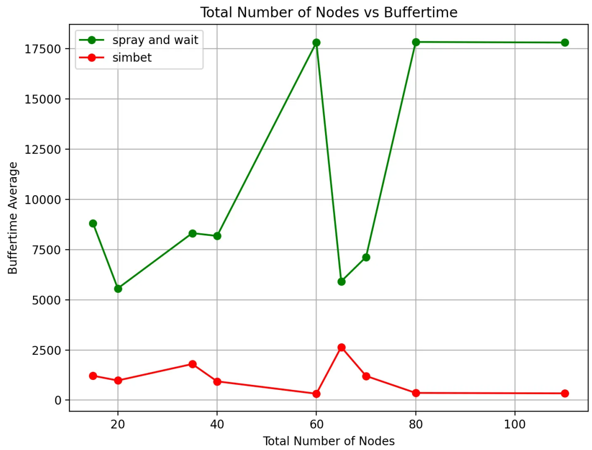

Figure 4.4: A plot showing the average end-to-end latency achieved within the scenario for each scale of the network.

When investigating the average buffer time for the Spray and Wait protocol, an opposite trend is observed, as shown in Figure 4.5b. This supports the hypothesis that there are more redundant packets throughout the network due to the higher time spent within the relay node buffer's. Although the overall latency of the network is low, the average buffer-time has an upwards trend. This is because even though more potential paths makes the presence of a short successful path highly likely several redundant paths are created which suffer from large storage delays. This is influenced by the intermittent connectivity of the network and the high mobility of the nodes [4]. A more selective protocol, such as SimBet, is able to take advantage of the more favourable connections more effectively [8]. This may become a more appropriate protocol for larger networks, as the high use of the limited buffer space can result in congestion and data loss if these trends continue [4].

Although a large overhead is expected, due to the redundant packets, it was found that SimBet had a significantly higher overhead ratio - as shown in Figure 4.5a. This is likely due to the implementation of SimBet that has been chosen, which may be inefficient within the network conditions. This has produced longer paths with a greater number of retransmissions relative to its successful deliveries [5].




(a) Trends in average overhead.




(b) Trends in average buffer time.

Figure 4.5: A visualisation of statistics gathered from the generated reports for each scale of the network.

# Chapter 5

# Wider Discussions

The transmission of data between nodes must rely on DTNs, when no traditional communication link can be formed. Although this is essential for highly mobile and remote networks, such as wildlife tracking, there are also popular uses for networks that require minimal data loss, such as disaster or emergency scenarios.

## 5.1 Wildlife Monitoring

The development of smart oceans is currently a widely researched topic and has yet to be implemented significantly. However, around the world, DTNs are often used for wildlife tracking and data collection to monitor the population of species and behaviour trends.

An example of this is the work by Wang et. al in 2004 [13], who deployed 'ZebraNet' to collect movement traces of zebra groups within Kenya. Each animal was equipped with a collar that contained a solar battery, GPS, and radio transmitter. Large importance had been placed on the energy-efficiency of the mobile network, which had meant that they were unable to receive constant data due to preserving the system in-between long battery charges. The ZebraNet Project allowed for the testing of mobile sensor networks capabilities with GPS sensors and aided studies on migration patterns and the interaction with other animal species [13]. Due to the available technology at the time, this study was limited by the co-operation of the animals wearing the sensor collars. Since then miniaturised wearable sensors have been develop to aid in the collection of important data while minimising the discomfort brought to the animal.

The decline in wildlife populations around the world is a growing issue that people of all vocations are striving to fix. Within BBC Science Focus Magazine [14] , DR Claire Asher brings much needed attention towards this crisis and highlights the recent advances in technology being made within this domain. ICARUS, the International Cooperation for Animal Research Using Space, are building from the existing radio-tracking technologies, discussed within our project, to develop a large global scale network. This will take advantage of the available miniaturised wearable sensors and the existing dense satellite network, in order to track wildlife populations around the world. This project aims to increase our understanding of animal behaviour, such as their communication and navigation abilities, as well as their movement patterns on a wide scale. Dr Asher continues to explain the importance that this otherwise unreachable data will have on reinstating our wildlife populations, exploring the impact the changing climate has on their behaviour, as well as the hopes ICARUS has of developing early detection of natural disasters and disease outbreaks [14].

This advancement in technology will largely improve the current systems used to track and monitor wildlife, which rely on purpose-built nearby infrastructure or tag-retrieval to collect the sensor data [12]. It can be said that these difficulties have held back research, due to economic feasibility and sparse or hard to access areas — such as ocean or artic environments.

## 5.2 Interplanetary Communications

In order to create these interconnected networks throughout a smart ocean, there are many considerations that are also present within wider scenarios such as interplanetary communications. As the original motivation for delay tolerant networks, data transmission within a space environment suffers from communication across large distances with intermittent communication and thus experience large delays and are prone to errors [4]. These communication issues are largely the result of node mobility. However, interplanetary communication differs from most applications due to the predictability of connection loss. The frequent disconnection within the network can be forecast due to the orbital motion of the planets and round trip times, however, disruptions due to available resources and error rates can be difficult to prevent [15]. The use of DTNs presents an alternate paradigm that allows for successful communication within these environments, which would not be possible following only traditional networking techniques [16].

## 5.3 Audio Multimedia Streaming

The use of delay tolerant networking following multi-hop propagation does not only benefit scenarios that do not have a clear end-to-end connection between the source and destination. Scenarios that instead suffer from long or variable delays will also struggle at sustaining efficient communication using traditional techniques. This is especially the case for applications that require quickly received acknowledgments for transmitted packets, or those that are greatly affected by the loss of data [4]

In situations such as streaming audio, there is a high need for reliable transmissions since frequent dropped packets have a large impact on user's end experience. This is less forgiving than video transmission, which can be extrapolated to replace missing data. It is significantly harder to reliably recover packets within audio multimedia streaming, since high packet loss results in unintelligible audio and may completely prevent communication between users. In order to redirect importance on the end-user's experience, prioritisation must be placed on minimising the packet loss rather than creating an efficient network with reduced delays [17].

A similar ideology can be used within our own scenario, as smart oceans suffer from highly expensive transmission of data, therefore, it is important to prioritise data loss to prevent costly re-transmissions. However, due to the nature of the data being collected, it is likely that the records can be extrapolated to replace any missing data points and minimise the use of expensive resources.

# Chapter 6

# Conclusion

The potential applications of sensor networks are ever growing along with the advancements being made within technology, the ongoing trend continues to shrink wearable sensors while increasing their performance capabilities [14]. This opens up extensions to the existing 'Internet of Things', by allowing the creation of large scale delay tolerant networks to facilitate the sharing of data among non-traditional links [1]. In particular, this aids in the retrieval of otherwise unattainable knowledge and furthers the understanding of our planet [14].

Often, great importance is placed on deciding the protocols within these mobile networks, as the methodology chosen can often have drastic changes to the available performance. The choice between a replication based or single-copy protocol can have large impacts on resource efficiency, both energy usage and congestion of node buffers, as discussed within the work of Daly [2]. The nature of the scenario has a large impact on the most suitable protocol, as there must be considerations of whether priority should be placed on minimising delays or maximising the reliability of packet delivery.

Within the smart-ocean-based scenario it was found that the scale of a network also has a significant impact on the performance of a replication-based protocol compared to a single-copy protocol. From the carried out experiments, it is clear that use of the spray and wait methodology is more preferable for greater populated networks. For our scenario, this was seen to achieve a higher ratio of successful deliveries, which are made across shorter paths, compared to that produced from the SimBet routing method. However, this comes with a large increase in overall network traffic and use of the limited resources available. Due to the nature of our scenario, a real world implementation is likely to favour a protocol that limits these redundant transmissions, since the cost of data transfer and storage is incredibly high within an underwater network [1]. It is likely that more research would be needed on real-world data, in order to confirm the advantages of message passing with an overall lower redundancy compared to that with longer routing paths.

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